

Supplementary Material for “Exact Bayesian Inference for Spike-and-Slab Priors in Takagi–Sugeno–Kang Fuzzy Systems with Approximately Calibrated Model-Averaged Prediction Intervals”

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Notation and method definitions follow the main text (Sections 2–3). All TSK methods use $R = 5$ rules from fuzzy c -means ($m = 2$) with Gaussian membership spreads equal to the per-cluster standard deviation of training points scaled by $s = 1.5$; results are averaged over 30 independent 80/20 train–test splits with a fixed seed unless noted otherwise.

1. Full MAE Values

Table 1: MAE comparison across all methods and targets (mean $\pm$ std, 30 splits). Best per target in **bold**.

Method	Energy-Heating	Energy-Cooling	Concrete
TSK-LS	1.32 $\pm$ 0.13	1.79 $\pm$ 0.13	6.65 $\pm$ 0.36
Bayesian-TSK	1.32 $\pm$ 0.13	1.79 $\pm$ 0.13	6.64 $\pm$ 0.36
TSK-SpikeSlab-BIC	15.72 $\pm$ 2.98	18.24 $\pm$ 2.74	43.16 $\pm$ 4.00
TSK-SpikeSlab-Gibbs	1.32 $\pm$ 0.13	1.78 $\pm$ 0.13	6.62 $\pm$ 0.36
Random Forest	0.33$\pm$0.03	1.02 $\pm$ 0.08	3.61$\pm$0.24
SVR	1.80 $\pm$ 0.16	2.05 $\pm$ 0.16	7.77 $\pm$ 0.33
Gaussian Process	0.34 $\pm$ 0.03	0.92$\pm$0.06	4.01 $\pm$ 0.25

The MAE ordering mirrors the R^2 ordering of the main text: the three dense Bayesian/least-squares TSK methods are effectively tied, the BIC-based approximation collapses, and the non-interpretable references (Random Forest and Gaussian Process) attain the lowest MAE.

2. Slab Variance Sensitivity (Coefficient-Level SSVS)

Table 2 reports the coefficient-level SSVS ablation on the Energy-Cooling target (10 splits, $\pi = 0.5$, $R = 5$ rules). Varying the slab variance τ^2 from 0.1 to 100 (three orders of magnitude)

spans the full sparsity–accuracy trade-off: a tight slab ($\tau^2 = 0.1$) over-shrinks the coefficients and collapses accuracy to $R^2 = -5.42$, whereas a diffuse slab ($\tau^2 = 100$) merely recovers the dense baseline ($R^2 = 0.92$ versus 0.94 for TSK-LS) while pruning nothing. No τ^2 setting improves on the dense baseline.

Table 2: Coefficient-level SSVS τ^2 ablation on Energy-Cooling (10 splits). Dense TSK-LS reference: $R^2 = 0.94$.

τ^2	R^2	RMSE	PICP	MPIW
0.1	−5.42	24.11	1.00	94.57
0.3	−1.57	15.04	0.96	59.42
1.0	0.88	3.24	0.93	12.55
3.0	0.91	2.90	0.90	11.03
10.0	0.91	2.85	0.89	10.76
100.0	0.92	2.62	0.91	9.88

3. Irrelevant-Feature Ablation

Table 3 reports the irrelevant-feature regime on the Energy-Cooling target (10 splits): up to 30 standard-normal noise features are appended to the $d = 8$ inputs. The dense TSK-LS degrades only modestly (from $R^2 = 0.93$ to 0.88, where the 0.93 is the 10-split dense baseline), and neither rule-level (TSK-SpikeSlab-Gibbs) nor coefficient-level (TSK-SSVS) sparsity improves upon it. Correct fuzzy c -means already provides robustness to irrelevant features, so sparsity does not improve on the dense baseline in this low-dimensional regime.

Table 3: Irrelevant-feature ablation on Energy-Cooling (R^2 ; dense TSK-LS vs rule-level Gibbs vs coefficient-level SSVS).

Appended noise features	TSK-LS	TSK-SpikeSlab-Gibbs	TSK-SSVS
0	0.932	0.932	0.884
4	0.945	0.934	0.881
12	0.878	0.878	0.876
30	0.875	0.875	0.834

4. Rule-Count Sensitivity

Table 4 sweeps the rule count R from 3 to 10 on all three targets (10 splits). Two conclusions follow. First, $R = 5$ is a reasonable default: the dense (TSK-LS) and conjugate (Bayesian-TSK)

accuracy saturates by $R \approx 5-7$ on every target, and the Gibbs sampler tracks the dense baseline at every R , providing a consistency check of the sampler outside its default configuration. Second, the sparsity boundary is robust to the rule count: even at $R = 10$ (an over-parameterized model with $R(d+1) = 90$ consequent coefficients) the Gibbs sampler keeps nearly all rules active (mean $9.7/10$), so rule-level sparsity does not activate even when the model is heavily over-parameterized. The BIC-based approximation (TSK-SpikeSlab-BIC) collapses to negative R^2 at every rule count.

Table 4: Rule-count sweep (R^2 , mean over 10 splits). “Rules” is the mean number of active rules of the Gibbs sampler (dense methods always keep R).

R	Energy-Heating					Energy-Cooling					Concrete				
	LS	Bayes	BIC	Gibbs	Rules	LS	Bayes	BIC	Gibbs	Rules	LS	Bayes	BIC	Gibbs	Rules
3	0.949	0.948	-2.17	0.948	2.6	0.918	0.917	-12.88	0.916	2.7	0.611	0.625	-5.34	0.673	3.0
4	0.983	0.979	-0.47	0.976	4.0	0.950	0.946	-1.51	0.939	4.0	0.714	0.717	-6.09	0.729	4.0
5	0.970	0.970	-1.37	0.970	4.9	0.932	0.932	-2.53	0.932	4.9	0.747	0.747	-5.82	0.747	5.0
7	0.967	0.967	-7.16	0.966	6.6	0.929	0.928	-9.71	0.927	6.6	0.750	0.748	-6.40	0.748	7.0
10	0.976	0.975	-5.79	0.973	9.7	0.939	0.939	-8.82	0.934	9.7	0.758	0.773	-2.34	0.777	9.7

5. Ablation Isolating the Failure of the BIC Baseline

Table 5 isolates which step of the BIC-plus-Laplace baseline causes its collapse on Energy-Cooling (10 splits, $R = 5$). Four configurations are compared:

1. **BIC weights + BMA** – all 2^R rule subsets are weighted by $\exp(-\text{BIC}/2)$ and averaged, so no hard threshold and no Laplace covariance are used;
2. **BIC-PIP + threshold + Laplace** – the per-rule BIC inclusion probabilities are hard-thresholded at 0.5, the selected model is fit by least squares, and the Laplace predictive covariance is used;
3. **Gibbs-PIP + threshold + Laplace** – the same prediction as (2), but the inclusion probabilities come from the Gibbs sampler rather than BIC;
4. **Full BIC baseline** – the as-implemented baseline of the main text, which additionally fits the selected model with ridge regularization.

Three conclusions follow. First, averaging is the decisive change: BIC weights fed through BMA (row 1) restore dense accuracy and near-calibration ($R^2 = 0.95$, PICP 0.92) even though

the model probabilities still come from BIC, whereas hard-thresholding those same probabilities to a single model (row 2) clearly underperforms ($R^2 = 0.60$). Second, the Gibbs inclusion probabilities are reliable: hard-thresholding and Laplace prediction built on them (row 3) reach the dense baseline ($R^2 = 0.94$, PICP 0.94). Third, the as-implemented baseline (row 4) collapses to $R^2 = -2.5$, with part of the gap attributable to its ridge-regularized least-squares fit of the highly collinear TSK design matrix (condition number $\sim 10^{18}$). Gibbs sampling with BMA is the only configuration that is simultaneously accurate and calibrated.

Table 5: Ablation of the BIC-plus-Laplace failure mechanism on Energy-Cooling (mean over 10 splits, $R = 5$). Dense TSK-LS reference: $R^2 = 0.94$.

Configuration	R^2	PICP	MPIW
BIC weights + BMA (no threshold)	0.95	0.92	7.8
BIC-PIP + threshold + Laplace (LS)	0.60	0.93	26.1
Gibbs-PIP + threshold + Laplace (LS)	0.94	0.94	8.4
Full BIC baseline (as implemented)	-2.53	0.00	9.9

6. MCMC Convergence Diagnostics

Table 6 reports multi-chain convergence diagnostics for the rule-level Gibbs sampler (TSK-SpikeSlab-Gibbs) on a representative split of two targets. Three chains are run with independent seeds (42, 123, 456), each with 1,000 burn-in and 2,000 retained draws. The Gelman–Rubin statistic satisfies $\hat{R} \leq 1.001$ for the consequents (β) and the noise variance (σ^2), and the effective sample size exceeds 1,700 for σ^2 , indicating well-mixed β and σ^2 chains. The inclusion-indicator chains are (near-)constant on these benchmarks—posterior inclusion probability saturates near 1.0 for every rule—so their effective sample size equals the chain length by construction rather than from active exploration; on Concrete the γ chain is exactly constant. Diagnostics are reported for the rule-level sampler; on these benchmarks the γ chains of the coefficient-level SSVS also saturate near full activation, so we expect qualitatively similar mixing, but we do not claim formal convergence verification for those runs.

Table 6: MCMC convergence diagnostics for TSK-SpikeSlab-Gibbs (three chains, 1,000 burn-in, 2,000 retained draws). $\hat{R}$ is the Gelman–Rubin statistic; ESS is the effective sample size.

Target	$\hat{R}(\gamma)$ (max)	$\hat{R}(\beta)$ (median / max)	$\hat{R}(\sigma^2)$	ESS(σ^2)
Energy-Cooling	1.000	1.000 / 1.001	1.000	1,725
Concrete	constant (= 1.0)	1.000 / 1.001	1.000	1,744

7. High-Dimensional Sparsity Boundary

Table 7 tests the sparsity boundary on a real high-dimensional regression benchmark (Superconductivity, UCI id 464; subsample of $n = 3000$ rows with $d = 81$ features, 30 splits, $R = 5$ rules). The pattern of the low-dimensional benchmarks reproduces almost exactly: rule-level sparsity (TSK-SpikeSlab-Gibbs) matches the dense baselines in R^2 (0.794 versus 0.792–0.795) while keeping calibration at the nominal level (PICP 0.940), and coefficient-level sparsity (TSK-SSVS) over-shrinks across the full slab-variance grid: Table 7 reports the two settings that bracket the low-dimensional sweep of Table 2 ($\tau^2 = 1$, collapsing R^2 to 0.674; $\tau^2 = 10$, recovering only to 0.763, still below dense), and a 10-split sweep of the complete grid confirms the pattern: R^2 rises monotonically from 0.36 at $\tau^2 = 0.1$ to 0.79 at $\tau^2 = 100$, no setting improves on the dense baseline (0.792), and the most diffuse slab ($\tau^2 = 100$) merely recovers it (0.794). The sparsity boundary therefore is not a low-dimensional artifact and does not depend on a particular slab-variance choice; whether sparsity is beneficial under different antecedent structures or much larger samples remains open. On this high-dimensional target, the rule-level Gibbs sampler also maintains near-nominal calibration (PICP 0.940) where the dense conjugate baseline slightly undercovers (PICP 0.921), consistent with the calibration contribution of the main text.

Table 7: High-dimensional sparsity boundary on Superconductivity (mean $\pm$ std over 30 splits; subsample $n = 3000$, $d = 81$, $R = 5$). PICP is the 95% interval coverage; MPIW is the mean prediction-interval width.

Method	RMSE	R^2	PICP	MPIW
TSK-LS (dense)	15.46 $\pm$ 0.82	+0.792 $\pm$ 0.024	–	–
Bayesian-TSK (dense conjugate)	15.33 $\pm$ 0.77	+0.795 $\pm$ 0.023	0.921	55.6
TSK-SpikeSlab-Gibbs	15.37 $\pm$ 0.70	+0.794 $\pm$ 0.021	0.940	59.7
TSK-SSVS ($\tau^2 = 1$)	19.35 $\pm$ 0.56	+0.674 $\pm$ 0.018	0.940	75.1
TSK-SSVS ($\tau^2 = 10$)	16.48 $\pm$ 0.55	+0.763 $\pm$ 0.018	0.935	65.0

8. Mean Prediction-Interval Width

Table 8 reports the mean prediction-interval width (MPIW) for the three probabilistic methods across all targets (30 splits). The compact widths support the calibration claim of the main text (Section 5.3, $\text{MPIW} \approx 6.7\text{--}32.9$).

Table 8: MPIW for the probabilistic methods (mean over 30 splits).

Method	Energy-Heating	Energy-Cooling	Concrete
Bayesian-TSK	6.61	9.20	32.04
TSK-SpikeSlab-Gibbs	6.69	9.40	32.87
Gaussian Process	1.75	5.18	20.46

9. Synthetic Validation of the Inclusion Full Conditional

The fixed-activation correctness check of the main text (Section 3.3) exercises the β and σ^2 updates but not the inclusion full conditional of the main text (the γ_j update of Section 3.4), because the indicators are fixed there. On the real benchmarks the indicators also saturate near full activation. We therefore validate the γ_j update on a synthetic problem with *known sparse* ground truth, where the exact posterior over rule configurations is computable.

Setup. We generate $n = 600$ samples from a synthetic TSK system with $R = 5$ rules and $d = 2$ features, in which only rules 0 and 2 have nonzero consequents (ground truth active set $\{0, 2\}$), with additive Gaussian noise of variance $\sigma^2 = 1$. The fuzzy partition is built by fuzzy c -means as in the main text. We then (i) run the rule-level block-Gibbs sampler ($\pi = 0.5$, $\tau^2 = 10$, $\sigma^2 \sim \text{InvGamma}(0.01, 0.01)$, 2,000 burn-in, 10,000 retained draws) and (ii) compute the *exact* posterior inclusion probabilities by enumerating all $2^5 = 32$ rule configurations, evaluating each configuration’s marginal likelihood in closed form via the Woodbury identities, and integrating σ^2 out over its inverse-gamma prior.

Result. Table 9 reports the sampler’s inclusion probabilities against the exact values. The Gibbs sampler recovers the known ground truth: rules 0 and 2 are active with probability ≈ 1.0 , and the inactive rules are pruned (inclusion probability ≈ 0.004), matching the exact posterior to a maximum absolute difference of 8×10^{-4} . The BMA predictive distribution also agrees with the exact model-averaged predictive (maximum absolute mean difference 0.006; median relative

predictive-variance difference 2.5%). This confirms the arithmetic of the inclusion full conditional and the BMA construction against exact computation, complementing the fixed-activation check of the main text.

Table 9: Synthetic validation of the γ_j update of the main text: posterior inclusion probabilities, Gibbs sampler vs exact model averaging over all 2^R configurations ($n = 600$, $d = 2$, $R = 5$; ground-truth active rules $\{0, 2\}$).

Rule	0	1	2	3	4
Gibbs	1.0000	0.0044	1.0000	0.0017	0.0034
Exact	1.0000	0.0044	1.0000	0.0019	0.0026
Gibbs–Exact	0.0000	0.0000	0.0000	0.0002	0.0008